

Data Mining Final Exam Questions and Answers

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Question 1

Based on the following table, calculate the Gini index and Gain Ratio for the feature credit_rating.

RID	age	income	student	credit_rating	Class: buys_computer
1	youth	high	no	fair	no
2	youth	high	no	excellent	no
3	middle_aged	high	no	fair	yes
4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle_aged	low	yes	excellent	yes
8	youth	medium	no	fair	no
9	youth	low	yes	fair	yes
10	senior	medium	yes	fair	yes
11	youth	medium	yes	excellent	yes
12	middle_aged	medium	no	excellent	yes
13	middle_aged	high	yes	fair	yes
14	senior	medium	no	excellent	no

Answer:

First, we obtain the total entropy of the dataset. We have 14 records, 9 of which are yes and 5 of which are no, so the entropy is calculated as follows:

$$H(D) = -\frac{9}{14} \log_2 \left(\frac{9}{14} \right) - \frac{5}{14} \log_2 \left(\frac{5}{14} \right) = 0.9403$$

Next, we calculate the Info of the feature credit_rating. This feature has two values, fair and excellent, for which the distribution of the target feature is as follows:

- **fair:** 8 records (6 records yes and 2 records no)

- **excellent:** 6 records (3 records yes and 3 records no)

Calculating the entropy for each value:

$$I(6, 2) = -\frac{6}{8} \log_2 \left(\frac{6}{8} \right) - \frac{2}{8} \log_2 \left(\frac{2}{8} \right) = 0.8113$$

$$I(3, 3) = -\frac{3}{6} \log_2 \left(\frac{3}{6} \right) - \frac{3}{6} \log_2 \left(\frac{3}{6} \right) = 1.0$$

The information amount (Info) for this feature is:

$$\begin{aligned} \text{Info}_{\text{credit_rating}}(D) &= \frac{8}{14} \times I(6, 2) + \frac{6}{14} \times I(3, 3) \\ &= \frac{8}{14}(0.8113) + \frac{6}{14}(1) = 0.8922 \end{aligned}$$

In the next step, we compute the Gain:

$$\text{Gain} = H(D) - \text{Info}_{\text{credit_rating}}(D) = 0.9403 - 0.8922 = 0.0481$$

Then we obtain the Split Info:

$$\text{SplitInfo} = -\frac{8}{14} \log_2 \left(\frac{8}{14} \right) - \frac{6}{14} \log_2 \left(\frac{6}{14} \right) = 0.9852$$

And finally, the Gain Ratio is calculated:

$$\text{Gain Ratio} = \frac{\text{Gain}}{\text{SplitInfo}} = \frac{0.0481}{0.9852} = 0.0488$$

Calculating the Gini Index:

$$\text{Gini}(D) = 1 - \left(\frac{9}{14} \right)^2 - \left(\frac{5}{14} \right)^2 = 0.459$$

$$\text{Gini}(\text{fair}) = 1 - \left(\frac{6}{8} \right)^2 - \left(\frac{2}{8} \right)^2 = 1 - \frac{36}{64} - \frac{4}{64} = \frac{24}{64} = 0.375$$

$$\text{Gini}(\text{excellent}) = 1 - \left(\frac{3}{6} \right)^2 - \left(\frac{3}{6} \right)^2 = 1 - \frac{1}{4} - \frac{1}{4} = 0.5$$

$$\text{Gini}_{\text{split}}(\text{credit_rating}) = \frac{8}{14}(0.375) + \frac{6}{14}(0.5) \approx 0.428$$

Question 2

Considering the actual and predicted classes in the adjacent table, calculate the Accuracy, Precision, Recall, and Specificity metrics.

Instance ID	Actual Label	Predicted Label
1	+	+
2	+	+
3	-	+
4	-	-
5	+	+
6	+	-
7	-	+

Answer:

First, we draw the Confusion Matrix for this model:

		Predicted	
		+	-
Actual	+	TP = 3	FN = 1
	-	FP = 2	TN = 1

Now we derive the metrics:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = \frac{4}{7} \approx 0.5714$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{3}{5} = 0.6$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{3}{4} = 0.75$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} = \frac{1}{3} \approx 0.3333$$

Question 3

Using weighted KNN (k-nearest neighbors with $k = 3$ and Euclidean distance), find the label of the test sample ($A = 1, B = 1, C = 1$).

Answer:

As can be seen, in our dataset for the first instance, the exact record in question, namely ($A = 1, B = 1, C = 1$), already exists, and its class is positive (+). Since this

record is already present in the dataset, there is no need to recalculate distances, and its class is directly known. Therefore, the predicted class for this sample will be **positive (+)**.

Question 4

a) If in a 4-class classification problem, we train a decision tree model and the model's accuracy considering the AUC metric is 35 percent, is this model suitable for use in the AdaBoost algorithm? Explain this.

b) Suppose we have 7 data samples and the output of classifier M in the first step of the AdaBoost algorithm is as shown in the table below. Then, calculate the error of M , the weight of the model in the current step, and the updated weight of each sample.

Instance ID	Actual Label	Predicted Label
1	+	-
2	+	+
3	-	-
4	-	+
5	+	-
6	+	+
7	-	-

Answer a:

As we know, AdaBoost is an algorithm based on the boosting method. In this method, the accuracy of the base (weak) model must be at least slightly better than a random guess. In a balanced classification, this value is equivalent to $AUC \approx 0.5$. Since the accuracy of this model is $AUC = 0.35$, its performance is even worse than random guessing. Therefore, this model by itself is not a suitable choice for the algorithm's base, unless it is first improved using methods to meet the minimum accuracy condition.

Answer b:

First, an equal initial weight is considered for all samples:

$$W_i = \frac{1}{7} \approx 0.1428$$

According to the table, the predictions for samples **1**, **4**, and **5** were incorrect (3 errors

out of 7 samples). Thus, the weighted error for the first round is:

$$err_1 = \frac{3}{7} \approx 0.4286$$

Since the error value is less than 0.5, the loop continues. The Classifier Weight is calculated according to the following formula:

$$\alpha = \frac{1}{2} \ln \left(\frac{1 - err}{err} \right) = \frac{1}{2} \ln \left(\frac{1 - \frac{3}{7}}{\frac{3}{7}} \right) = \frac{1}{2} \ln \left(\frac{\frac{4}{7}}{\frac{3}{7}} \right) = \frac{1}{2} \ln \left(\frac{4}{3} \right) \approx 0.1438$$

Next, we update the weights. For correctly predicted samples:

$$W_{\text{correct}} \leftarrow W_i e^{-\alpha} = \left(\frac{1}{7} \right) \times (0.4286)^{-0.14} \approx 0.12$$

For incorrectly predicted samples:

$$W_{\text{incorrect}} \leftarrow W_i e^{\alpha} = \left(\frac{1}{7} \right) \times (0.4286)^{0.14} \approx 0.16$$

For normalization, we first calculate the sum of the new weights ($S = 3(0.16) + 4(0.12) = 0.96$):

- Final weight for incorrect data:

$$W_{\text{incorrect_final}} = \frac{0.16}{0.96} \approx 0.166$$

- Final weight for correct data:

$$W_{\text{correct_final}} = \frac{0.12}{0.96} = 0.125$$